SCHOOL OF TECHNOLOGY

Final Presentation for KCA Connect Agentic AI

Presenter: IGNATIUS LUMOSI

Reg No.: 22/05647

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Supervisor: GRIFFIN KENGA

Project Overview

- KCA Connect Agentic AI is an intelligent conversational assistant designed specifically for KCA University. The system serves as a unified communication platform connecting students, faculty, and administrators through an AI-powered interface.

- By integrating academic resources and institutional data, the platform delivers real-time, accurate information while leveraging cutting-edge artificial intelligence to transform how the university community accesses and shares information.

Background & Context

Fragmented Channels

KCA University currently relies on disconnected communication methods including emails, office visits, and group chats that lack centralization.

Slow Information Flow

Information sharing is often delayed, failing to meet the immediate needs of students and staff.

AI Opportunity

Recent advances in artificial intelligence present a compelling opportunity to automate administrative support and streamline communication.

Centralized Solution

Digital assistants offer a proven path to creating a unified information hub that serves the entire university community effectively.

Problem Statement

Inefficient Communication

Current communication channels may result in delays in information delivery across the institution.

Missing Critical Information

Students also face frustration due to inaccessible or missing information regarding timetables, fee structures, and academic requirements.

Existing Process Limitations

Existing delivery methods cannot adequately scale to meet the needs of a large and growing university population.

System Fragmentation

Disconnected systems cause frequent errors and confusion among both learners and administrative staff, reducing overall institutional efficiency.

Solution

Contextual Retrieval

Leveraging Retrieval-Augmented Generation (RAG) with a Qdrant vector database to pull precise, context-aware information from institutional data.

Multi-LLM Orchestration

Intelligent routing via LangChain across Gemini, Groq, and Cerebras Large Language Models for optimal response quality and reliability with fallback.

Grounded, No Hallucination

All answers are firmly rooted in verified institutional documents, ensuring accuracy and completely eliminating AI hallucinations for trusted information.

Secure User Experience

A premium React frontend web dashboard, secured with JWT authentication and powered by a FastAPI backend and Supabase for seamless, streaming chat interactions.

System Objectives

Primary Objective

To develop an AI-powered digital assistant for KCA University.

Real-Time Information Access

Provide instantaneous(immediate response) access to academic and administrative information including timetables, fees, course details, and institutional policies.

Communication Efficiency

Improve communication efficiency between students and staff by eliminating intermediaries and providing direct centralized access to verified information.

Error Reduction

Minimize delays and errors caused by manual information delivery through automated, consistent, and accurate responses to routine queries.

Scope

This project encompasses a defined set of technical and operational boundaries for the KCA Connect Agentic AI, ensuring focused development and clear functionality.

1. Backend Architecture

- FastAPI-based server with multi-LLM fallback routing (Gemini, Groq, Cerebras)

2. Knowledge Base

- Semantic indexing of institutional PDFs, DOCX, and XLSX files
- Qdrant Vector Database for efficient retrieval

3. Frontend Dashboard

- Secure user management and admin tools
- Knowledge ingestion interface for staff

4. Security & Governance

- JWT-based authorization
- Campus-specific content filtering
- Role-based access control

Significance

Real time information delivery

- Our solutions ensures that all the information required by any member of the university community is provided to them when they need it as they will just ask .

Centralized information management

- The system ensure the information required by members is at a single access point unlike the current disconnected methods

Secure, Appealing and easy to use web system for users

- Use of JWT for token validation ensure all sessions are verified
- Visually appealing front end designed using the tailwind css for the glass effects
- Fast interactive system and is easy to use

Literature Review

Research conducted demonstrated the transformative potential of AI-powered assistants in higher education institutions globally.

Holmes et al. (2019)

AI-powered assistants enhance communication efficiency and automate routine processes, significantly reducing administrative burden while improving response times.

Deakin University's 'Genie'

Implemented AI assistant improved operational efficiency and student satisfaction through 24/7 support availability and instant query resolution.

Georgia State's 'Pounce'

Chatbot deployment reduced enrollment delays, increased student engagement, and demonstrated measurable improvements in retention rates.

Case for KCA: KCA Connect scales for a localized solution tailored specifically for Kenyan academic structures(KCA University) and institutional requirements.

Methodology

Requirements Collection
Structured interviews, questionnaires and web search was used to research and for data collection to ground KCA Connect Agentic AI in real university workflows and eliminate generic responses.

Agentic Development Model
Agile Scrum supported iterative delivery of the agentic architecture, including RAG, semantic search, and multi-LLM routing, with each sprint validated against KCA’s real data and feedback.

Technical Stack
Built with a Python backend, Fast API, Lang Chain orchestration, Qdrant vector database, React 18, Supabase, JWT security, and multi-provider fallback to ensure fast, reliable, and secure responses.

Validation Testing
Rigorous functional and performance testing achieved 100% pass rate, 2.45-second latency, and 95% accuracy, confirming dependable grounding in institutional data for production deployment.

Results and achievements

- ✓ Successfully developed a fully functional Kca connect agentic ai system which after verification of running proved to be a success
- ✓ KCA Connect Agentic AI has achieved significant milestones, demonstrating robust functional success, high performance, and impactful operational improvements.

100% Functional Success

Pass rate across 14 critical test cases, with all core features validated and production-ready.

2.45s Performance Metrics

Average response latency, exceeding the 5-second target. Achieved 95% accuracy in information retrieval and successfully ingested pages of institutional data.

Tested System Health

Handled complex Excel fee structures and multi-format documents, with reliable multi-LLM fallback routing tested.

Conclusion

KCA Connect Agentic AI represents a significant advancement in how KCA University delivers information services. By leveraging artificial intelligence and modern development practices, the system addresses critical communication challenges while positioning the institution as a technology leader.

The project demonstrates practical application of software engineering principles to solve real-world institutional problems, providing valuable experience in AI integration, system design, and stakeholder collaboration.

This implementation serves as a foundation for future enhancements and potential expansion to additional services and integrations across the university ecosystem.

Thank You

Live System Demonstration